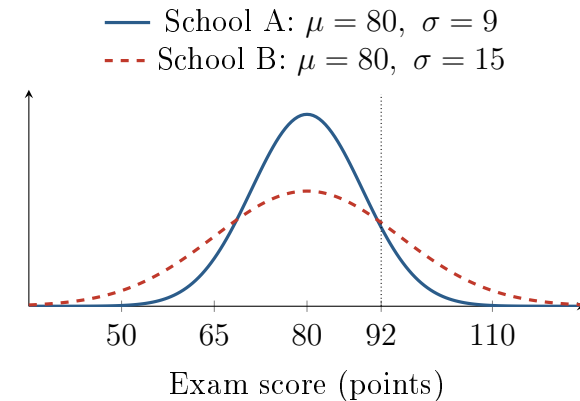


Comparing a Score of 92

Unit 2 · Topic 2.11 · THE PROBLEM

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE — one sentence before you work the parts

Would a 92 stand out equally at both schools? Give one reason from the picture.

- DOK 1** (a) Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.
- DOK 2** (b) For a score of 92, calculate the z -score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.
- DOK 3** (c) In 3–4 sentences, decide whose claim is better supported. Use both percentiles from (b) and explain how the different spreads change what a score of 92 means at each school.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.